

- Ecol.* **131**, 155–171 (1997).
13. Wills, C. & Condit, R. Similar non-random processes maintain diversity in two tropical rainforests. *Proc. R. Soc. Lond. B* **266**, 1445–1452 (1999).
 14. Penfold, G. C. & Lamb, D. Species co-existence in an Australian subtropical rain forest: evidence for compensatory mortality. *J. Ecol.* **87**, 316–329 (1999).
 15. Webb, C. O. & Peart, D. R. Seedling density dependence promotes coexistence of Bornean rain forest trees. *Ecology* **80**, 2006–2017 (1999).
 16. Wills, C., Condit, R., Foster, R. B. & Hubbell, S. P. Strong density- and diversity-related effects help to maintain tree species diversity in a neotropical forest. *Proc. Natl Acad. Sci. USA* **94**, 1252–1257 (1997).
 17. Augsburger, C. K. Seedling survival of tropical tree species: interactions of dispersal distance, light gaps, and pathogens. *Ecology* **65**, 1705–1712 (1984).
 18. Gilbert, G. S., Hubbell, S. P. & Foster, R. B. Density and distance to adult effects of a canker disease of trees in moist tropical forest. *Oecologia* **98**, 100–108 (1994).
 19. Cintra, R. A test of the Janzen–Connell model with two common tree species in Amazonian forest. *J. Trop. Ecol.* **8**, 529–536 (1997).
 20. Carson, W. P. & Root, R. R. Herbivory and plant species coexistence: community regulation by an outbreaking phytophagous insect. *Ecol. Monogr.* **70**, 73–99 (2000).
 21. Packer, A. & Clay, K. Soil pathogens and spatial patterns of seedling mortality in a temperate tree. *Nature* **404**, 278–281 (2000).
 22. Streng, D. R., Glitzenstein, J. S. & Harcombe, P. A. Woody seedling dynamics in an east Texas floodplain forest. *Ecol. Monogr.* **59**, 177–204 (1989).
 23. Jones, R. H., Sharitz, R. R., Dixon, P. M., Segal, D. S. & Schneider, R. L. Woody plant regeneration in four floodplain forests. *Ecol. Monogr.* **64**, 345–367 (1994).
 24. Connell, J. H. Diversity in tropical rain forests and coral reefs. *Science* **199**, 1302–1310 (1978).
 25. Hubbell, S. P. Seed predation and the coexistence of tree species in tropical forests. *Oikos* **35**, 214–229 (1980).
 26. McCanny, S. J. Alternatives in parent–offspring relationships in plants. *Oikos* **45**, 148–149 (1985).
 27. Hubbell, S. P., Condit, R. & Foster, R. B. Presence and absence of density dependence in a neotropical tree community. *Phil. Trans. R. Soc. Lond. B* **330**, 269–281 (1990).
 28. Schupp, E. W. The Janzen–Connell model for tropical tree diversity: population implications and the importance of spatial scale. *Am. Nat.* **140**, 526–530 (1992).
 29. Clark, D. A. & Clark, D. B. Spacing dynamics of a tropical rain forest tree: evaluation of the Janzen–Connell model. *Am. Nat.* **124**, 769–788 (1984).
 30. Clark, J. S., Macklin, E. & Wood, L. Stages and spatial scales of recruitment limitation in southern Appalachian forests. *Ecol. Monogr.* **68**, 213–235 (1998).

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Geographic structure and dynamics of coevolutionary selection

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Coevolution of species is one of the major processes organizing the Earth's biodiversity. Recent coevolutionary theory has indicated that the geographic structure of species has the potential to impose powerful and continuing effects on coevolutionary dynamics, if that structure creates selection mosaics and coevolutionary hotspots across landscapes^{1–7}. Here we confirm that current coevolutionary selection in interspecific interactions can be highly divergent across both narrow and broad geographic scales, thereby fuelling continuing coevolution of taxa. Study of a widespread plant–insect interaction across a broad range of habitats for several years showed that an insect functioning both as a pollinator and a floral parasite can be strongly

mutualistic in some habitats but commensal or antagonistic in neighbouring habitats. The results for one of the habitats span seven years, demonstrating that the local structure of coevolutionary selection can remain stable across multiple generations. Conservation of the evolutionary processes maintaining long-term biological diversity may require preservation of the conditions that allow a long-term shifting geographic mosaic of coevolutionary hotspots and coldspots.

Although some coevolving interactions show little spatial variation in outcome⁸, evidence for pronounced geographic differences in interspecific interactions has now been found in a diverse array of interactions, including those between parasites and hosts, predators and prey, competitors, and mutualists^{9–18}. The geographic mosaic theory of coevolution indicates that these geographic differences are an inherent part of the coevolutionary process, driven by variation between habitats in the trajectories of natural selection (selection mosaics), the occurrence of reciprocal selection in only some communities (coevolutionary hotspots), and a constantly changing

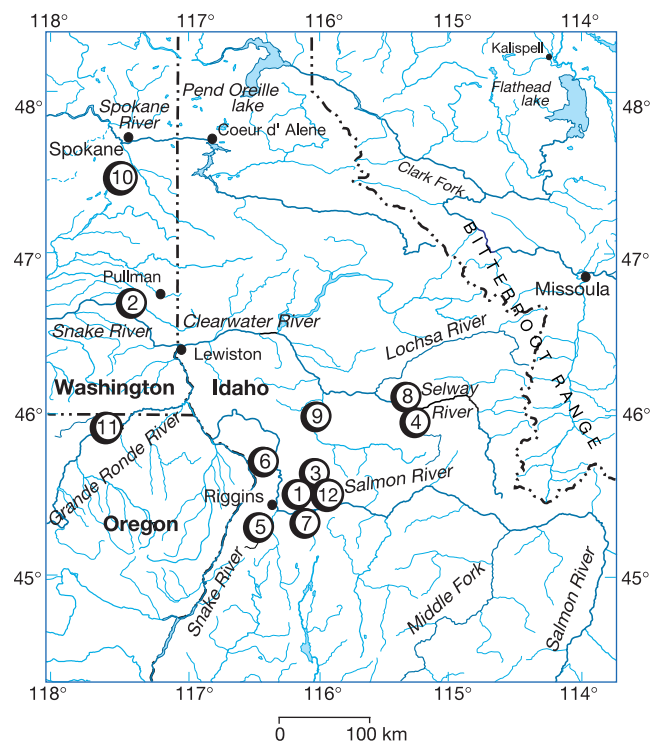


Figure 1 Distribution of study sites for analysis of coevolutionary hotspots in the interaction between the plant *Lithophragma parviflorum* and the moth *Greya politella*. Locations: 1, Berg = Berg Mountain, Salmon River, Idaho (ponderosa pine woodland); 2, Granite = Granite Point, Snake River, Washington (canyon bottom meadow steppe); 3, Keating = Keating Ridge, above Salmon River, Idaho (ridgetop grassland with ponderosa pine); 4, Meadow = Meadow Creek near confluence with Selway River, Idaho (streamside meadow amid grand fir and western red cedar); 5, Rapid = Rapid River above confluence with Little Salmon River, Idaho (streamside narrow canyon amid Douglas fir and Pacific yew); 6, Saddle = near Pittsburg Saddle above Snake River, Idaho (high-elevation open ridge top with scattered ponderosa pines); 7, Salmon = along Salmon River east of Riggins, Idaho (north-facing steppe); 8, Selway = lower Selway River, Idaho (openings in habitat dominated by western red cedar, Douglas fir and ponderosa pine); 9, South Fork = South Fork of Clearwater River, Idaho (open ponderosa pine savanna); 10, Turnbull = Turnbull National Wildlife Refuge, Washington (ponderosa pine woodland on glacial polygonal soil); 11, Wenaha = Wenaha River, Idaho (south-facing steppe with thick shrub cover); 12, Wind = Gospel Hump Wilderness along Salmon River, Idaho (dry, south-facing steppe).

genetic landscape shaped by gene flow and other evolutionary forces^{1,19}. Although coevolutionary hotspots and coldspots have now been demonstrated in a few interactions on the basis of the geographic distribution of coevolved traits^{11,12}, the distribution of current coevolutionary selection is unknown for any interaction. The absence of such data precludes the development of more ecologically grounded models of coevolutionary dynamics.

We evaluated current coevolutionary selection in the interaction between the herbaceous plant *Lithophragma parviflorum* (Saxifragaceae) and the floral-parasitic moth *Greya politella*^{20,21}. This is one of the most widespread plant–insect interactions in western North America and is pervasive in the Rocky Mountains, northern USA, a floristically diverse region that includes large expanses of relatively pristine environments shaped by rugged topography and Pleistocene events^{22,23}. Extensive field surveys between 1994 and 1999 within the major river drainages of western Idaho and adjacent areas of Washington and Oregon showed that all sampled *L. parviflorum* populations support *Greya* moth populations (Table 1). Moreover, the interaction is common within populations, with more than 25% of plants receiving eggs in most populations and years. At Turnbull National Wildlife Refuge in Washington State, where the local *L. parviflorum* population includes hundreds of thousands of individuals, 85–98% of plants received eggs during three years of study.

In all surveyed habitats, moths oviposited into developing flowers by inserting the ovipositor down the corolla and into the floral ovary. The moths are very close relatives of yucca moths²⁴ and, like those moths, pollinate their obligately outcrossing host flowers during oviposition. Unlike the yucca moths, however, pollination by *Greya* is passive rather than active, and occurs as pollen adhering to the abdomen touches the stigma during oviposition. Because the moths leave direct evidence of their visits (that is, eggs within flowers), we were able to assess the outcome of this pairwise interaction, even when the same plants were visited by co-pollinators, by comparing flowers with and without *Greya* eggs. Consequently, unlike most interactions, it was possible to evaluate this pairwise interaction within the full community context in which it occurs in natural populations.

In most surveyed habitats, the moths were restricted to feeding on *L. parviflorum*, both as larvae and as adults. Hence, the interaction is inherently beneficial to the moths in these communities. However, previous work^{20,21} and the preliminary observations during the surveys indicated that the interaction is at least sometimes less obligate for the plants, because co-pollinators visit the plants in some habitats, thereby potentially reducing *Lithophragma*'s reliance on *Greya* for pollination. Unlike the co-pollinators, the larvae eat a small subset of the hundreds of developing seeds within each capsule. Hence, the overall outcome of the interaction has the potential to be mutualistic, commensal or antagonistic, depending on the presence and abundance of effective co-pollinators in different habitats.

Table 2 Effect of moth oviposition on probability of floral development in *L. parviflorum*

Population	Effect of <i>Greya</i> oviposition on floral development		
	1997	1998	1999
Mutualistic effect of <i>Greya</i>			
Berg	1.1	3.0	1.8
Meadow	—	2.4	1.6
Saddle	0.8	20.5	9.2
Turnbull	1.0	1.0	2.4
Antagonistic effect of <i>Greya</i>			
Rapid	0.5	0.9	1.3
Salmon	—	0.4	—
South Fork	—	0.6	0.4
Wenaha	2.1	0.4	1.6
No effect of <i>Greya</i>			
Granite	0.5	1.1	1.0
Keating	1.1	1.1	1.2
Selway	—	0.9	—
Wind	—	0.7	1.0

Results are shown as the ratio of the percentage of developed capsules containing eggs to the percentage of aborted capsules containing eggs. Significant effects ($P < 0.05$) determined by χ^2 tests are shown in bold. Values significantly >1 indicate selective development of capsules containing eggs. Values significantly <1 indicate selective abortion of capsules containing eggs. The number of dissected capsules for each analysis averaged 678 (s.d. = 89.5, range = 386–898). Populations indicated with a dash were either not studied in that year or had too few aborted capsules for statistical analysis. Analyses were restricted to floral positions 2–4 on each plant.

We evaluated whether *L. parviflorum* relied upon *Greya* for pollination in 12 contrasting habitats ($N = 18,912$ flowers in total), ranging from dry open steppe to ponderosa pine (*Pinus ponderosa*) woodlands to steep basaltic or granitic slopes along moist river drainages (Fig. 1). In all except one of the habitats (Rapid River), the moths depend completely on *L. parviflorum* for adult and larval food and for mating sites. In each community we tested whether flowers visited by the moths were more likely to develop seeds than those not visited by the moths.

The interaction varied among habitats from mutualism to commensalism to antagonism. *Lithophragma parviflorum* relied on *G. politella* for pollination within four of the habitats in one or more years (Table 2). These four habitats provide evidence for current mutualistic hotspots: communities in which the moths depend completely on this one host plant and in which the plants depend upon the moths for pollination. Developed capsules at these sites were on average 1.8–20.5-fold more likely to have *Greya* eggs than were aborted capsules. The most extreme dependence of *Lithophragma* on *Greya* for pollination occurred at Saddle, near the Seven Devils Wilderness, Idaho. Although the plants in that habitat aborted 59–69% of their flowers in 1998 and 1999, they aborted only 2–4% of flowers containing *Greya* eggs. Developed capsules were 20.5-fold more likely to have *Greya* eggs than aborted capsules in 1998 and 9.2-fold more likely in 1999 (both $\chi^2 P < 0.0001$).

Within four other habitats, *Greya* pollination and oviposition had no effect on floral development, indicating a commensal relationship in those populations (Table 2). A previous four-year

Table 1 Percentage of *Lithophragma parviflorum* plants with *Greya politella* eggs at sites within and near the northern Rocky Mountains, USA

River system	Plants with eggs (%)	Range over years	Total no. of plants sampled	No. of years sampled
Snake River, Washington	27	11–41	572	3
Turnbull NWR, Washington	93	85–98	599	3
Salmon River, Idaho				
Uplands	39	6–68	1,193	3
Lowlands	30	18–39	1,195	4
Clearwater River, Idaho	40	33–47	399	2
Selway River, Idaho	45	27–58	1,228	5
Rapid River, Idaho	66	36–49	348	4
Weiser River, Idaho	12	—	406	1
Snake River, Idaho	52	33–62	559	3
Wenaha River, Oregon	26	21–32	500	3

An average of 197 plants were sampled for each population in each year. All plants were separated by at least 1 m. Three capsules on floral positions 2–4 were dissected from each plant.

study using a different experimental protocol at one of these sites (Granite Point, Washington) demonstrated that abundant co-pollinators swamp the mutualistic effects of *G. politella* in that steppe habitat, reducing the outcome to commensalism^{20,21}. The earlier results (1987–90) together with the additional three years reported here a decade later (1997–99) indicate that the interaction at Granite Point remained commensal over multiple years.

The remaining four habitats showed evidence of an antagonistic relationship between the plants and the moths. Plants selectively aborted capsules containing *Greya* eggs, indicating current antagonistic coevolutionary hotspots (Table 2). Aborted capsules at these sites were on average 2.2-fold (range 1.6–2.4-fold between sites) more likely to have *Greya* eggs than developed capsules. For example, along the South Fork of the Clearwater River in Idaho, plants aborted 25% of flowers in both 1998 and 1999, but they aborted 34–38% of flowers containing *Greya* larvae.

These results indicate that the outcome of the interaction between *Greya* and *Lithophragma* differs significantly between habitats. We evaluated the outcome further by testing whether developing capsules visited by *Greya* had more or fewer seeds per flower than flowers not visited by the moths. Developing flowers with eggs did not differ significantly in mean seed number from flowers on the same plant without eggs at most sites and years. Only 4 of 33 comparisons between habitats and years (12 habitats for 2–3 years) showed significant differences in the mean number of seeds per flower, 2 in 1997 and 2 in 1999 (two-way analyses of variance or Wilcoxon signed-rank tests with $P < 0.05$ criterion; $N = 769,723$ seeds). Hence, the major effect of *Greya* is on whether capsules develop at all rather than on the number of seeds that mature within a capsule. Flowers not visited by *Greya* at the mutualistic sites often simply fail to develop through lack of pollination, whereas flowers at the antagonistic and commensal sites are sufficiently pollinated by co-pollinators for the positive effects of *Greya* on pollination to be swamped.

We quantified the relative rarity of co-pollinators at one of the mutualistic sites, Turnbull National Wildlife Refuge in Washington State, in 1999. *Greya* was the most common visitor, composing 42% of visits ($N = 1,146$ visits by all floral visitors). Another 32% of visits were by reliable co-pollinators, classified as reliable according to the experimental results of an earlier study²¹. The co-pollinators included several bombyliid fly and bee species^{20,21}. The remaining 26% were by insects whose visits rarely result in effective pollination, as determined by earlier studies²¹.

Visitation rate per flower was low. Averaged over the season, flowers had a probability of 0.47 of being visited by *Greya* on each of the one to several days for which they were open, but a probability of only 0.36 of being visited by a co-pollinator. Because only some visits by *Greya* and co-pollinators result in pollination²¹, these low visitation rates indicate that some flowers abscise through lack of pollination after the 1–2 days for which each flower remains open. Hence, pollination of flowers at Turnbull in 1999 relied strongly on visits by *Greya*, corroborating the results obtained in the capsule dissections (Table 2).

These overall results confirm a major prediction from recent coevolutionary theory: that geographically structured species will tend to coevolve toward a complex spatial mosaic of coevolutionary hotspots and coldspots. Although this is the most extensive analysis undertaken so far on the geographic structure of current phenotypic selection acting on species interactions across natural populations, it covers only part of the geographic range of this interspecific interaction. However, additional complex outcomes are likely to occur in other habitats, depending on the community context of the interaction. The results therefore indicate two general points about the geographic mosaic of coevolution and its effects on the organization of biodiversity. First, coevolution in geographically structured species may result in complex patterns of phenotypic selection among habitats. Recent work on other interspecific interactions has

shown differences in coevolved traits across broad geographic landscapes^{9–18}. The differences in current coevolutionary selection across even smaller-scale geographic landscapes in *Greya* and *Lithophragma* indicate that the long-term ebb and flow of coevolutionary selection might be important sources of ecological and evolutionary dynamics in natural biological communities. Second, if coevolving interactions commonly display selection mosaics and a mix of coevolutionary hotspots and coldspots, then severe anthropogenic fragmentation of landscapes is likely to have major effects on how continuing coevolution contributes to the persistence and dynamics of the Earth's biodiversity. □

Methods

Field surveys

Field surveys on the geographic occurrence of the interaction were conducted in the major river drainages of Idaho and adjacent Oregon and Washington by collecting the second, third and fourth capsules produced on plants distributed along linear transects running parallel to the river ($N = 6,999$ plants in total). All plants were separated by at least 1 m from all other plants. Capsules were placed in ethanol, returned to the laboratory and dissected under a microscope to determine the presence of *Greya* eggs or larvae.

Evaluation of outcome of the interaction

Although females also take nectar from the flowers, significant pollination by *Greya* occurs only during oviposition²⁵. Within each population we evaluated whether the second to the fourth flowers on each plant were more likely, or less likely, to develop mature seeds when they contained *Greya* eggs. Each plant generally produces four to eight flowers in sequence along a floral stalk. We chose the second to the fourth flowers because previous experiments had shown that plants rarely suffer resource limitation when maturing seeds within the first four flowers that they produce²¹. Hence, floral abortion through resource limitation is unlikely to affect these early flowers. The three-year analysis included the dissection of 18,912 floral capsules, including both developed capsules (that is, those with mature seeds) and aborted capsules (that is, those producing no mature seeds and withering on the plant).

We tested whether developed capsules visited by *Greya* had more or fewer seeds per flower than flowers not visited by the moths by comparing mean seed numbers for pairs of developed capsules: one with *Greya* eggs and one without (the mean number of floral pairs compared per habitat each year was 56.6; the total number of maturing seeds counted was 769,723). Analysis was restricted to randomly chosen pairs of second to fourth flowers taken from individual plants.

Quantification of pollinator visitation rates

We quantified pollinator visitation rates at Turnbull National Wildlife Refuge by recording all visits to *L. parviflorum* flowers within randomly chosen plots during 72 h of observation distributed across the peak week of the flowering season in 1999 ($N = 1,146$ visits by floral visitors). Each 1 m × 1 m plot contained multiple flowering individuals. Each observer recorded all visits to a plot within 1 h and then moved to a new plot.

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1. Thompson, J. N. *The Coevolutionary Process* (Univ. Chicago Press, Chicago, 1994).
2. Hochberg, M. E. & van Baalen, M. Antagonistic coevolution over productivity gradients. *Am. Nat.* **152**, 620–634 (1998).
3. Nuismer, S. L., Thompson, J. N. & Gomulkiewicz, R. Gene flow and geographically structured coevolution. *Proc. R. Soc. Lond. B* **266**, 605–609 (1999).
4. Case, T. J. & Taper, M. L. Interspecific competition, environmental gradients, gene flow, and the coevolution of species' borders. *Am. Nat.* **155**, 583–605 (2000).
5. Gomulkiewicz, R., Thompson, J. N., Holt, R. D., Nuismer, S. L. & Hochberg, M. E. Hot spots, cold spots, and the geographic mosaic theory of coevolution. *Am. Nat.* **156**, 156–174 (2000).
6. Hochberg, M. E., Gomulkiewicz, R., Holt, R. D. & Thompson, J. N. Weak sinks could cradle mutualisms—strong sources should harbour parasitic symbioses. *J. Evol. Biol.* **13**, 213–222 (2000).
7. Nuismer, S. L., Thompson, J. N. & Gomulkiewicz, R. Coevolutionary clines across selection mosaics. *Evolution* **54**, 1102–1115 (2000).
8. Holland, J. N. & Fleming, T. H. Geographic and population variation in pollinating seed-consuming interactions between senita cacti (*Lophocereus schottii*) and senita moths (*Upiga virescens*). *Oecologia* **121**, 405–410 (1999).
9. Berenbaum, M. & Zangerl, A. Chemical phenotype matching between a plant and its insect herbivore. *Proc. Natl Acad. Sci. USA* **95**, 13743–13748 (1998).
10. Benkman, C. W. The selection mosaic and diversifying coevolution between crossbills and lodgepole pine. *Am. Nat.* **153**, S75–S91 (1999).
11. Benkman, C. W., Holimon, W. C. & Smith, J. W. The influence of a competitor on the geographic mosaic of coevolution between crossbills and lodgepole pine. *Evolution* **55**, 282–294 (2001).
12. Brodie, E. D. III & Brodie, E. D. Jr Costs of exploiting poisonous prey: evolutionary trade-offs in a predator-prey arms race. *Evolution* **53**, 626–631 (1999).
13. Burdon, J. J. & Thrall, P. H. Spatial and temporal patterns in coevolving plant and pathogen associations. *Am. Nat.* **153**, S15–S33 (1999).
14. Kraaijeveld, A. R. & Godfray, H. C. J. Geographic patterns in the evolution of resistance and virulence in *Drosophila* and its parasitoids. *Am. Nat.* **153**, S61–S74 (1999).
15. Lively, C. M. Migration, virulence, and the geographic mosaic of adaptation by parasites. *Am. Nat.* **153**, S34–S47 (1999).
16. Radtkey, R. R., Fallon, S. M. & Case, T. J. Character displacement in some *Cnemidophorus* lizards revisited: A phylogenetic analysis. *Proc. Natl Acad. Sci. USA* **94**, 9740–9745 (1997).

17. Parker, M. A. Mutualism in metapopulations of legumes and rhizobia. *Am. Nat.* **153**, S48–S60 (1999).
18. De Jong, P. W., De Vos, H. & Nielsen, J. K. Demic structure and its relation with the distribution of an adaptive trait in Danish flea beetles. *Mol. Ecol.* **10**, 1323–1332 (2001).
19. Thompson, J. N. The evolution of species interactions. *Science* **284**, 2116–2118 (1999).
20. Thompson, J. N. & Pellmyr, O. Mutualism with pollinating seed parasites amid co-pollinators: constraints on specialization. *Ecology* **73**, 1780–1791 (1992).
21. Pellmyr, O. & Thompson, J. N. Sources of variation in pollinator contribution within a guild: the effects of plant and pollinator factors. *Oecologia* **107**, 595–604 (1996).
22. Caicco, S. L., Scott, J. M., Butterfield, B. & Csuti, B. A gap analysis of the management status of the vegetation of Idaho (U.S.A.). *Conserv. Biol.* **9**, 498–511 (1995).
23. Ricketts, T. H., Dinerstein, E., Olson, D. M. & Loucks, C. J. (eds) *Terrestrial Ecoregions of North America: A Conservation Assessment* (Island Press, Washington DC, 1999).
24. Pellmyr, O., Thompson, J. N., Brown, J. M. & Harrison, R. G. Evolution of pollination and mutualism in the yucca moth lineage. *Am. Nat.* **148**, 827–847 (1996).
25. Pellmyr, O. & Thompson, J. N. Multiple occurrences of mutualism in the yucca moth lineage. *Proc. Natl Acad. Sci. USA* **89**, 2927–2929 (1992).

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Spike train dynamics predicts theta-related phase precession in hippocampal pyramidal cells

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According to the temporal coding hypothesis¹, neurons encode information by the exact timing of spikes. An example of temporal coding is the hippocampal phase precession phenomenon, in which the timing of pyramidal cell spikes relative to the theta rhythm shows a unidirectional forward precession during spatial behaviour^{2,3}. Here we show that phase precession occurs in both spatial and non-spatial behaviours. We found that spike phase correlated with instantaneous discharge rate, and precessed unidirectionally at high rates, regardless of behaviour. The spatial phase precession phenomenon is therefore a manifestation of a more fundamental principle governing the timing of pyramidal cell discharge. We suggest that intrinsic properties of pyramidal cells have a key role in determining spike times, and that the interplay between the magnitude of dendritic excitation and rhythmic inhibition of the somatic region is responsible for the phase assignment of spikes^{4,5}.

Two competing frameworks have been suggested for the representation of information in the brain: rate coding and temporal coding. Rate coding suggests that information is represented by discharge rates over a group of neurons⁶. Temporal coding models, on the other hand, argue that the temporal relationship of spikes is more critical for information transfer^{7–11}. During spatial behaviours, hippocampal pyramidal neurons discharge specifically at

certain locations in the environment (the 'place field'), implying that hippocampal neurons 'code' for spatial information¹². Furthermore, phase assignment of spikes with respect to the theta rhythm shows a unidirectional advancement as the animal crosses the place field^{2,3}. Because spike phase shows a monotonic relation to position, whereas firing rate increases and decreases as the animal enters and leaves the place field, it was suggested that this 'temporal code' might be a reliable mechanism for representing space^{2,13}.

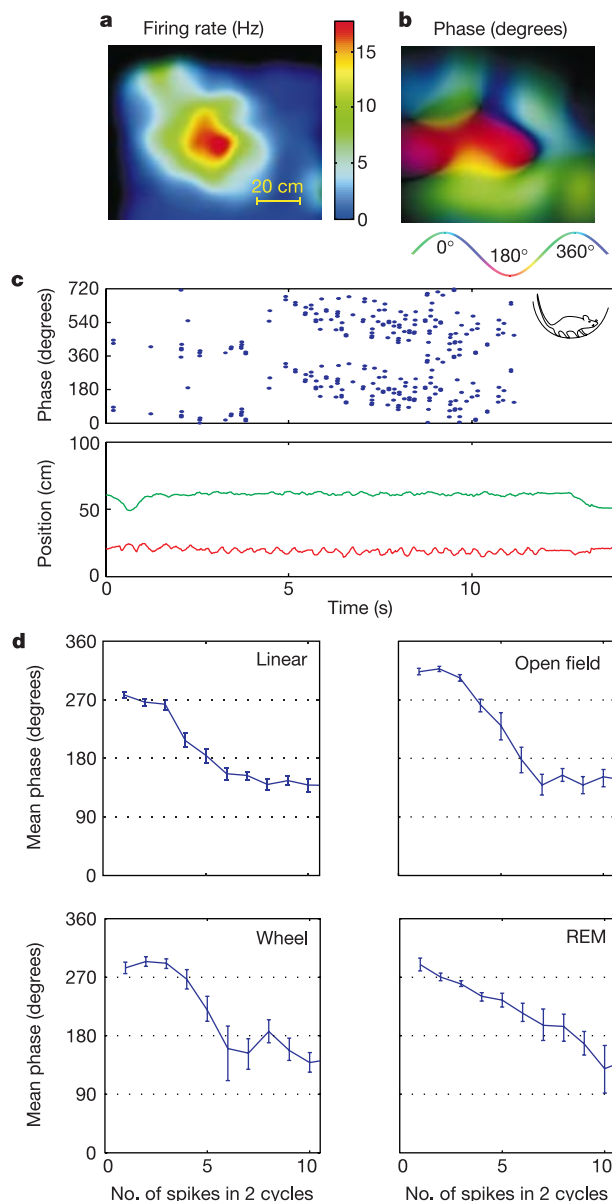


Figure 1 Relation between theta phase and instantaneous firing rate in pyramidal cells. **a**, Mean firing rate as a function of position, for an illustrative CA1 pyramidal cell ('place map'). **b**, Mean firing phase as a function of position for the same cell ('phase map'). **c**, Spike phase advancement during a wheel running episode. The cell fired sparsely at constant phase until 5 s into the trial, when an intense period of firing occurred, accompanied by phase precession, although the rat's head remained stationary (<3 cm). Top panel, phase and time of spikes. Bottom panel, instantaneous head position (red and green: X and Y coordinates). **d**, Rate versus phase relationship during four types of behaviour. For each spike, instantaneous rate was estimated by the number of spikes in the 720° interval centred on it. Graph shows circular mean and 95% confidence interval²³ of all spikes in the database, grouped by instantaneous rate. Phase 0 corresponds to the positive peak of CA1 pyramidal layer theta. The database consisted of 28, 181, 23 and 253 cells from 3, 5, 7 and 15 rats in the linear, open field, wheel and REM behaviours.